Social Media Engagement Analytics

Project Overview

This project focuses on analyzing engagement data from a social media dataset to uncover patterns in user behavior, content performance, and sentiment impact. The dataset contains metrics such as likes, comments, shares, impressions, watch time, follower count, and engagement rate.

Objective

To transform raw social media engagement data into actionable insights that help optimize content strategies, improve audience targeting, and enhance overall platform performance.

Tools Used

- **Python** (Google Colab / Jupyter Notebook)
- **Pandas & NumPy** — Data cleaning, wrangling, statistical analysis
- **Matplotlib & Seaborn** — Static visualizations
- **Plotly** — Interactive visualizations
- **CSV Dataset** — social_media_engagement_5000.csv

Project Workflow

1. **Data Ingestion**

Load the dataset into the environment for analysis.

- Import CSV using Pandas
- Verify file path and encoding
- Store data in a DataFrame

2. **Data Inspection**

Understand the dataset structure and metadata.

- Use head(), tail(), shape, info()
- Check column names and data types
- Identify categorical vs numeric fields

3. **Data Cleaning**

Correct errors and handle missing or duplicate values.

- Detect missing values with isnull()

- Apply dropna(), fillna(), or imputation
- Remove duplicates and standardize categories

4. **Data Transformation**

Convert and reshape data for analysis.

- Convert date columns to datetime
- Normalize or scale numeric fields
- Apply log transformations if needed

5. **Feature Engineering**

Create new variables to enrich analysis.

- Calculate engagement_score
- Extract hashtag counts
- Generate derived metrics like log_likes

6. **Exploratory Data Analysis (EDA)**

Explore distributions and relationships in the data.

- Generate descriptive statistics
- Use value_counts() for categories
- Create correlation matrix

7. **Statistical Analysis**

Quantify variability and central tendencies.

- Compute mean, median, mode
- Calculate variance and standard deviation
- Assess percentiles, skewness, kurtosis

8. **Data Visualization**

Present findings with charts and plots.

- Seaborn plots (count, bar, violin, heatmap)
- Matplotlib plots (scatter, line, histogram)
- Plotly interactive charts

9. Final Insights

Content Performance

- **Post Types**: **Reels** consistently show the **highest engagement** compared to images or text posts, indicating that short-form video content drives stronger interaction.
- **Content Category**: Categories like ***Travel* and *Food*** tend to **perform best**, attracting more likes and shares than others such as *News* or *General*.
- **Countries**: Engagement rates vary by geography, with countries like ***Brazil*, *India*, and *USA*** showing the **highest average engagement** levels.

User Trends

- **Age Impact**: **Younger users** (particularly those under 30) demonstrate **higher engagement rates**, while older age groups show lower interaction levels.
- **Verified Accounts: Verified accounts** generally **outperform** non-verified ones, with higher average engagement rates and stronger follower activity.

Behavioral Insights

- **Best Time of Day**: Impressions peak during **evening hours**, suggesting that posting later in the day maximizes visibility and reach.
- **Device Type**: **Mobile users** contribute the **longest** watch times, while desktop usage shows shorter engagement, highlighting the importance of mobile-optimized content.

Sentiment Analysis

- **Top Sentiment**: **Positive sentiment** posts **perform best**, generating higher engagement rates compared to neutral or negative ones.
- **Neutral/Negative Behavior**: **Neutral posts** tend to receive **moderate** engagement, while **negative sentiment** posts consistently **underperform,** showing lower likes, shares, and watch times.

10. Decision Making

Use insights to guide strategy and actions.

- Recommend optimal posting times
- Suggest focus on mobile-friendly content
- Emphasize positive sentiment for higher engagement

Conclusion

These insights reveal that **short-form video content, positive sentiment, and mobile-friendly strategies** are key drivers of engagement. Younger audiences and verified accounts amplify performance, while timing (evenings) and geographic targeting (Brazil, India, USA) further enhance results.